Findings - Round of Translation (NL ↔ EN)

Overview

We reviewed the first 150 subtitle-like lines produced by the translation model and scored each line on a 3-point scale (3 = correct, 2 = minor mistake, 1 = incorrect). Overall quality is mixed: everyday declaratives with simple syntax tend to be accurate, while conversational idioms, ellipsis, and game-show jargon cause the most errors. Roughly, about half of the lines are correct, a sizable portion contain minor issues that a human could fix in seconds, and a smaller but meaningful portion are incorrect enough to change the meaning.

Below I summarize the error patterns, explain why they occur from a linguistic point of view, give short examples (paraphrased), and list mitigation ideas.

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1) Lexical choice / word sense (polysemy)

What happens: The model selects a plausible dictionary sense but not the contextually correct one (e.g., *"plan"* as *"map"*; *"mate"* as *"partner/companion"* vs *"friend"*). Dutch–English pairs are rich in polysemy and false friends (e.g., *"proef"* ≠ *"proof"*; here it meant *"challenge/trial"*).
Why (linguistics): Word-sense disambiguation needs discourse cues and world knowledge. Subtitles are short, with little context, so senses are underspecified.
Impact: Often scored 2; sometimes 1 when the sense flips the scene (e.g., *"jury"* vs *"council"*).
Mitigation: Add domain glossaries (Survivor jargon: *tribal council, immunity idol/necklace, blindside, challenge*), train with in-domain subtitles, and use constrained decoding with a terminology list.

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2) Idioms & multiword expressions (MWE)

What happens: Literal translations of idioms: *"spill the beans"* → *"mors(t) de bonen"*; *"on the chopping block"* → literal cutting block; *"blindside him"* → *"overvallen/robbing"*.
Why: Idioms are non-compositional; token-level translation fails. Dutch also has idiomatic near-equivalents that differ in form.
Impact: Typically 1 (incorrect) because meaning is wrong even if grammar is fine.
Mitigation: Phrase-level alignment, idiom dictionaries, and MWE tagging during training; add synthetic paraphrases of frequent idioms.

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3) Pragmatics & register (discourse markers, fillers)

What happens: Over-literal handling of fillers and backchannels (*"yeah, yeah, yeah"*), vocatives, and imperatives. Elliptical speech gets normalized in unnatural ways, or punctuation is missing, changing tone.
Why: Subtitles compress speech; models trained on formal text mis-handle discourse particles (*"ja dus"*, *"weet je"*) and politeness markers.
Impact: Usually 2 (minor) unless it flips illocution (e.g., statement vs command).
Mitigation: Fine-tune on conversational corpora and subtitle data; keep a normalization layer that preserves pragmatic markers where appropriate.

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4) Tense, aspect, and modality

What happens: Errors with English progressive vs Dutch simple present (*"ik ben … aan het"*), with modality (*"zal/ga"* → *will/going to*), and with counterfactuals.
Why: Dutch periphrastic progressives map variably to English; deictic perspective (speaker's plan vs prediction) affects *will/going to*.
Impact: Often 2; occasionally 1 when time reference becomes wrong for the plot.
Mitigation: Add tense/aspect tags as features; include contrastive minimal pairs in fine-tuning.

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5) Negation & polarity

What happens: Dropped negation or polarity reversal in short clauses (*"I'm not going to…"* → *"I'm going to…"*).
Why: Subtitles use ellipsis and contractions; tokenization of *n't* and *geen/niet* is error-prone; long-distance negation scope is tricky.
Impact: High severity—score 1—because it flips meaning.
Mitigation: Use negation-aware data augmentation; enforce coverage checks for negation tokens at decode time.

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6) Word order & clause structure

What happens: Dutch V2/Vfinal structures are mapped to awkward or wrong English order; embedded clauses and relative clauses get scrambled.
Why: SOV-in-embedded vs SVO differences; separable verbs; subtitle fragments lacking overt subjects.
Impact: Readability drops (score 2) or meaning changes (score 1).
Mitigation: Syntactic pretraining (dependency objectives), or an intermediate reordering step for DE→EN.

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7) Named entities and show-specific terminology

What happens: Person names (*Locky, Ziggy, Jericho*) and proper nouns get inflected or replaced; *"tribal council"* becomes generic *"jury/council"*; *"immunity necklace"* → *"chain"*.
Why: Low-frequency tokens; subword splits; lack of domain exposure; confusing *jury* in Dutch reality-TV context.
Impact: Usually 2; sometimes 1 when role terms are swapped.
Mitigation: Copy-bias for entities, do-not-translate lists, and a Survivor glossary.

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8) Segmentation & truncation

What happens: Partial lines, duplicated phrases, or abrupt sentence cuts; sometimes the translation continues a previous line or stops early.
Why: Subtitle segments don't align perfectly; beam search with length penalties may truncate; punctuation missing in source.
Impact: Often 1 because content is incomplete.
Mitigation: Pre-align segments by timestamps; add sequence-boundary markers; increase min-length constraints; post-join short fragments before translation.

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9) Literal calques & false friends

What happens: *“proef” → “proof”*, *“immuniteit proef” → “immunity trial”*; *“verhaal”* as *“story”* where *“plan”/“pitch”* is meant; *“ondergang” → “kill him”*.
Why: Form similarity biases decoding; cross-lingual embedding proximity confuses near neighbors.
Impact: 1 or 2 depending on severity.
Mitigation: Counterexamples in training; lexical constraints for high-risk pairs.

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10) Coreference & pronouns

What happens: Wrong *he/she/they*; possessives mapped incorrectly (*“zijn/haar”*); dropped pronouns in fragments.
Why: Gender is not always overt in Dutch; zero anaphora and subtitle ellipsis increase ambiguity.
Impact: Meaning confusion in dialogue attribution (often 2).
Mitigation: Use document-level (multi-sentence) context; keep previous subtitle as additional input window.

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11) Morphology & compounding

What happens: Dutch compounds rendered word-by-word, or English compounds over-split; diminutives (*-je*) and aspectual particles mishandled.
Why: Subword tokenization lacks morphological awareness; compounds are productive in Dutch.
Impact: Usually 2 (fluency hit) unless semantics change (1).
Mitigation: Morphologically informed BPE/unigram models; compound splitter/merger in preprocessing.

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12) Punctuation, casing, and typographic conventions

What happens: Missing commas/periods; inconsistent casing on names; question marks omitted.
Why: Training data with noisy punctuation; subtitle constraints (line length) encourage drops.
Impact: Mostly 2 (readability).
Mitigation: Punctuation restoration model as a post-processor; truecasing.

What seems *easy* vs *hard*

- **Easy:** Short factual statements; present-tense SVO; copula clauses; common vocabulary.
- **Hard:** Idioms, slang, colloquialisms; strategic/game jargon; sentences with clause stacking; lines needing context from the previous subtitle; negation and modality.

Recommendations (actionable)

1. **Domain adaptation:** Fine-tune on reality-TV/subtitle corpora (OpenSubtitles + custom Survivor set).
2. **Terminology & entities:** Lock a glossary (e.g., *tribal council, immunity idol/necklace, blindside, challenge, vote-out*).
3. **Idiom handling:** Build an idiom lexicon; add parallel paraphrases; tag MWEs.
4. **Context window:** Feed previous/next subtitle to the model (document-level MT).
5. **Negation checks:** Add decoding constraints or a post-edit rule that flags polarity mismatches.
6. **Segmentation hygiene:** Merge very short fragments before translation; enforce min-length/coverage in decoding.
7. **Punctuation restoration:** Lightweight post-processor to fix casing and sentence ends.
8. **Human-in-the-loop:** Prioritize review on lines with idioms, negation, named entities, and show jargon (highest payoff).

Conclusion

Most errors are *explainable* via well-known linguistic challenges: non-compositional idioms, cross-lingual sense ambiguity, Dutch–English word order differences, pragmatic markers in conversation, and subtitle segmentation. The majority of problematic lines can be corrected with small, systematic interventions (terminology control, idiom lists, context, and punctuation restoration). Applying these targeted fixes plus in-domain fine-tuning should move many “2” scores to “3” and a good fraction of “1” scores to “2”, improving both adequacy and fluency for this domain.